[2. POSITION TOW-ARM CYLINDERS]

With the screed above the starting reference, adjust the height of the tow-points on both sides of the paver. The height of the tow-points affects the line of pull (traction angle) between the tow-point connection and the pivot point of the screed.

After the screed has heated, adjust both tow-points to the same height. Use the tow-point height controls located on the screed control boxes or on the operating consoles. There are two tow-point height scales. One is located adjacent to the tow-point connection. The other is above the tow-point connection at the front of the tractor.

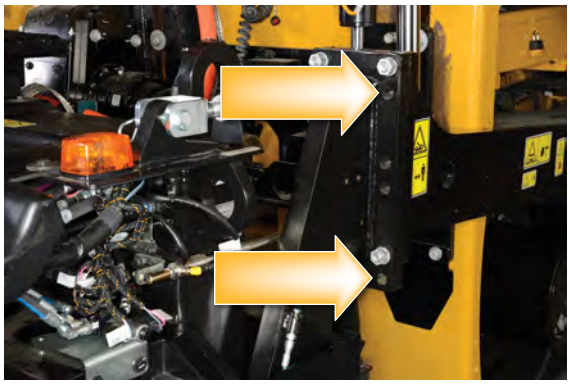
Align the pointer to the zero mark. Then raise the tow-point indicator above zero a distance equal to 1.5 times the thickness of the layer passing under the screed. For example, if the planned uncompacted layer thickness is 5 cm (2"), then position the pointer 7.5 cm (3") above the zero mark on the scale.



Set the tow-point height to create the correct line of pull.

User Tip: *The tow-point height scales use metric units. Each mark is two centimeters. Depending on the preferred units of measure, the scale can be converted to inches to avoid confusion and mistakes.*

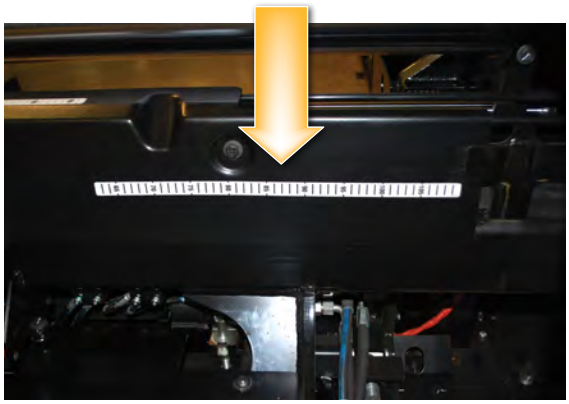
Cat Pavers can be set up to pave at deep depths. To facilitate deep-depth paving, reposition the tow-arm attachment on the drop arm to the bottom set of holes. Change the tow-arm position when paving between 18 cm (7") and 30 cm (12"). Then, set the tow-point height according to the paving depth.



Change the tow-arm position for deep-depth paving.

[3. SET THE PAVING WIDTH]

With the screed still above the starting reference, set the paving width. If utilizing a fixed width screed, extensions may need to be added, and the screed extensions will have to be bolted on. Typically, the screed will have hydraulic-power extenders. Using the scale that is facing the back of the screed, move the extender out to the desired setting. It is good practice to have equal extender on both sides of the screed, whenever possible. For example, if the paving width is 3.6 m (12') and the paver is equipped with a 3 m (10') screed, bring each extender out 30 cm (1').



Based on main screed width, set extender width according to the project requirements.

For some applications, the extender may be used to pave the emergency lane in addition to the driving lane. The emergency lane may have a different slope than the driving lane, and therefore, extender widths will be unequal for that application.

User Tip: *Remember the guidelines for adding auger extensions and mainframe extensions when planning the paving width. Add auger and mainframe extensions when paving with a front-mount screed to keep the end of the auger within 60 cm (2'), or closer, of the end gate. Add auger and mainframe extensions when paving with a rear-mount screed to keep the end of the auger within 1 m (3'), or closer, of the end gate.*

[4. SET MAIN SCREED CROWN]

Next, set the crown of the main screed according to project specifications. The crown indicator scale is located on the right side of the screed. Use the power crown controls on the right or left screed control panel to create the desired positive or negative crown. The crown should normally be set at zero. In other words, the screed is flat. The crown scale shows percent of slope from the center of the screed.



Some projects may require crown of the main screed.

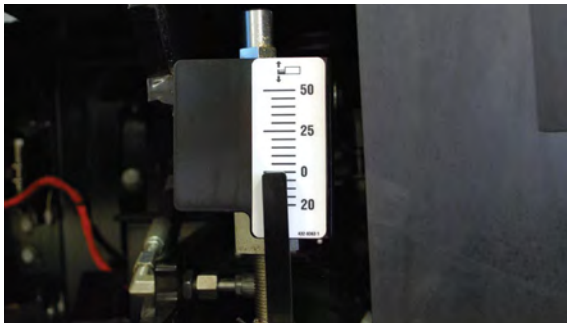
User Tip: *Caterpillar recommends verifying the accuracy of all scales on the screed. To set the crown scale, raise the screed and lock it in position. Use a string or straight edge to verify that there is no crown. Use the crown adjustment controls if it is necessary to flatten the screed. Then, check the crown scale. Move the indicator to zero. The crew should not have to "string the screed" at the beginning of every shift, if they periodically check the accuracy of the crown scale.*

[5. SET EXTENDER HEIGHT]

Screed extenders have adjustable height. Set the height of the extenders so they match the height of the main screed. When the height of the extenders is correct, there are no transition marks in the surface of the bituminous layer.

On Cat Screeds, there are height scales on the inner edge of each extender. Adjust extender height using controls on the right and left screed consoles. To help create the desired 5 mm (0.2") angle of attack in the main screed, set the height on both screed extenders 5 mm (0.2") above the zero mark on the scale.

Again, it is good practice to periodically verify that the height scales are calibrated to give an accurate measurement of extender height.



Set the extender height before lowering the screed to the starting reference.



[6. LOWER THE SCREED ON THE STARTING REFERENCE]

Before lowering the screed, raise the end gates to the proper position to ensure that the screed rests on the starting reference and not the end gates. Next, make sure the starting reference is correct. Use a starting reference with a thickness that matches the uncompacted mat depth that will be placed by the screed. Normally, the mat will compact about 6 mm (0.25 per inch) per 25 mm (1") of mat depth. For example, to achieve a 50 mm (2") compacted mat, use a starting reference that is 64 mm (2.5") thick. Use two references. Position them so they are under the pivot of the extender and completely supporting both the main screed and the screed extender from front to back.



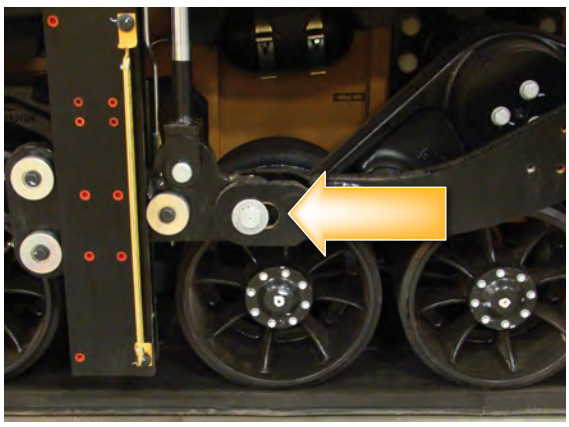
Boards should be placed at the outer edge of the main screed (behind the tractor) and outer edges of the extenders.

If starter boards are used, utilize at least two. Position the starter boards so they will support both the main screed and the extender. If a screed extender has been extended more than half its maximum width, use another starter board to support that extender.

User Tip: *Starter boards must be long enough to support the entire width of the main screed and the extender. For example, if both the main screed plate and the extender are 45 cm (18") long, then starter boards that are at least 90 cm (36") long will be needed.*

Lower the screed to the starting reference. The screed operator must be sure to leave the screed Raise / Lower switch in the "Lower" or "Float" position.

With the screed resting on the starting reference, move the paver ahead slightly to remove slack from the tow-point connection. The roller on the end of the tow arm must be in contact with the tow-point frame before the paver starts to pave. Removing the slack from the tow-point connection puts light tension on the screed frame, helping the screed float at the correct height when paving starts. If there is a gap between the tow-arm roller and the frame, the screed will dive off the starting reference.



Move the paver forward to remove slack from the tow-point connection.

[7. ADJUST THICKNESS SCREWS]

Thickness screws on both sides of the screed need to be adjusted to establish the angle of attack for the main screed. When laying down mats up to 10 cm (4") thick, adjust both screws so the indicators on the thickness control scales are aligned with zero. The main screed angle of attack will be approximately 6 mm (0.25") when the thickness screws are set at zero and the tow-point height is set at the correct distance above zero on the tow-point height scale.

When laying mats that are between 10 cm (4") and 30 cm (12"), adjust the screws to move the indicator progressively further into the plus range. Increasing the thickness-screw setting increases the main screed angle of attack. The thickness gauge must be at two marks positive or more for mats 20 cm (8") thick or thicker.



Thickness control scales are located on both sides of the screed.

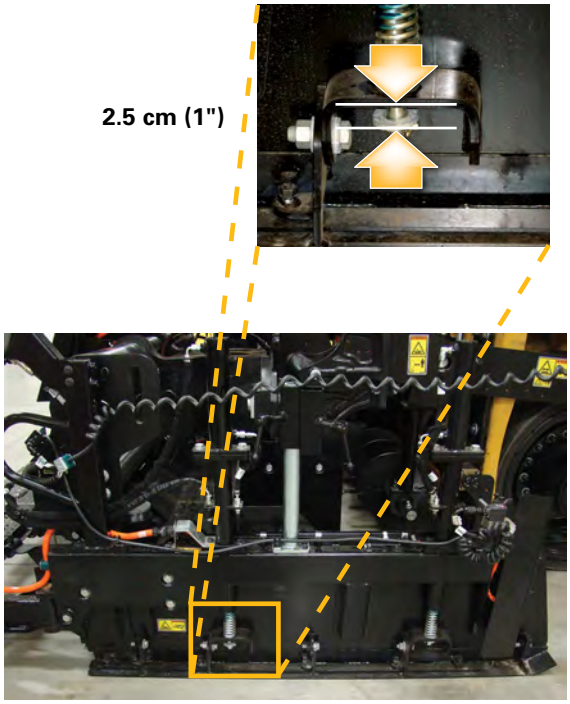
Note: *Thickness screw adjustment is dependent of mix type and other factors. The previously referenced numbers are a starting point and may have to be increased or decreased depending on mix type.*

[8. POSITION END GATES]

End gates on both ends of the screed retain material at the proper width. They float on the grade.

First, lower the end gate until it touches the grade. On manual gates, raise or lower the spring tension to create 2.5 cm (1") space between washer and bracket to allow the end gates to move up and down over any grade irregularities.

In certain situations, the end-gate ski may need to be positioned at the same height as the screed plate. For example, when creating a height match to an adjacent layer or to an existing gutter, adjust the end gate up to the same level as the screed plate.



Position the end gates in a manner that is appropriate for the application.

Note: *Always perform end-gate maintenance at the end of the shift, or more frequently when paving with polymer-modified asphalt cement. Pay particular attention to the slide area. It is a narrow opening where fines and aggregates often become trapped. The end gate is difficult to adjust when the slide area is dirty. Clean the slide area between the outer and inner frame members by completely lowering the end gate and spraying with an approved release agent.*

[9. SET AUGER HEIGHT]

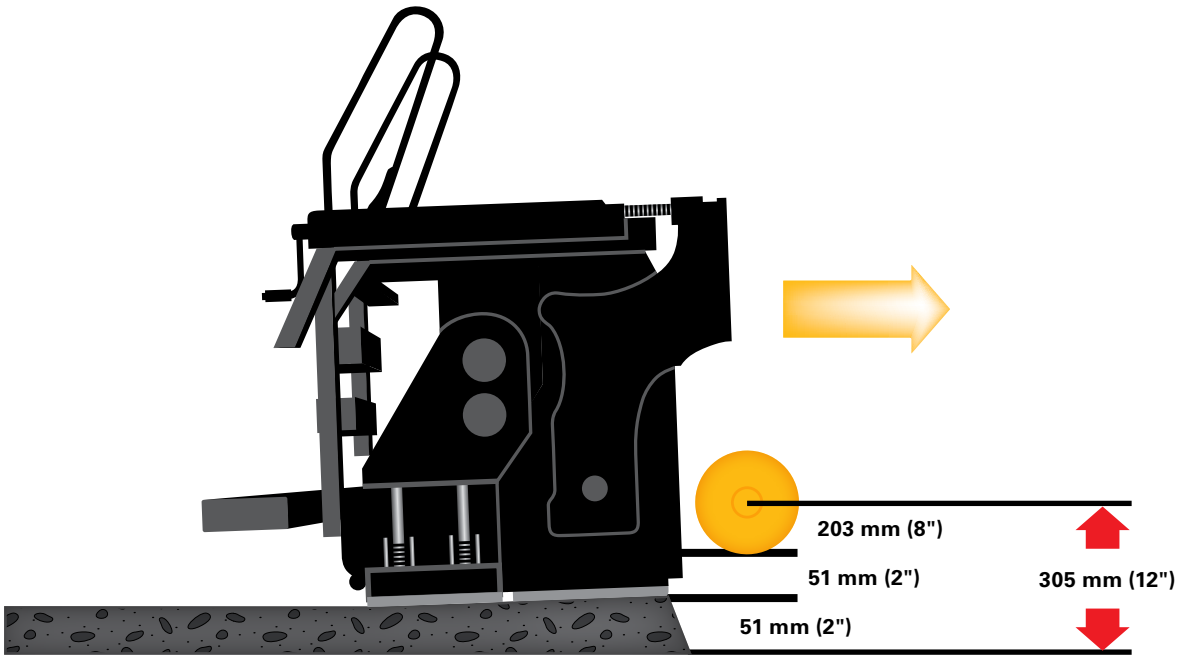
The height of the augers in relation to the depth of the uncompacted mat has an impact on the texture and finish of the mat.

If the augers are too low, open texture and material segregation may appear. If the augers are too high, the head of material will likely be too high, which will cause the screed to climb. Various mix designs react differently to auger height adjustment. As a rule, with aggregate size below 16 mm (0.6"), set the auger height at least 5 cm (2") above the height of the uncompacted mat. With aggregate size above 16 mm (0.6"), set the auger height at least 8 cm (3") above the height of the uncompacted mat.

The distance from the bottom of the auger to the center of the auger shaft is 200 mm (8"). Now add 50 mm (2") or 80 mm (3") to that. Then add the thickness of the mat to be paved. The total is the distance from the centerline of the auger shaft to the grade. Adjust the auger up or down until the dimension is reached.

If the mix is very coarse or very tender, fine-tune the auger height after paving starts.

The augers are too low, if open texture is evident in the layer directly in line with the auger shafts. Gradually increase the height of the augers until the texture of the bituminous layer is acceptable.



[10. POSITION FEEDER SENSORS]

Depending on user preference, Cat Pavers may be equipped with mechanical, paddle-type feeder sensors or with sonic feeder sensors. Both types must be positioned correctly for uniform material flow from the auger chamber to the end gates.

Paddle sensors contact material that is flowing out of the auger chamber. They are mounted to an adjustable shaft at the front of the auger chamber.

Position the paddle-type feeder sensors so they are detecting the active pile of material about 46 cm (18") away from the last auger segment. The paddle arm should be at a 45-degree angle at the 46 cm (18") distance.



Paddle sensors provide accurate control and are easy to adjust with a measuring device.

Position the sonic sensors installed on the end gates, so they point at the moving head of material that comes out the front of the auger chamber. Ideally, this target area should be about 45 cm (18") away from the end of the sonic sensor. The sensor should be positioned perpendicular to the target so the echoes from the sound pulses reflect back to the sensor.

It may be easier to make the final adjustments and position the sensors after paving has started when using sonic feeder sensors. Extend a measuring device out 45 cm (18") and put the measuring device along one side of the sensor to show exactly where the sensor is aimed. Reposition the sensor if necessary.



Sonic sensors should be positioned perpendicular to the moving head of material.

[11. ADJUST FEEDER SYSTEM CONTROLS]

Set the feeder control system before paving begins. Sometimes, it doesn't need to be adjusted when making a new pass—specifically, when making a new pass at the same width. However, if the paving width has changed or this is the first pass of the shift, start with the feeder controls at the recommended starting positions.

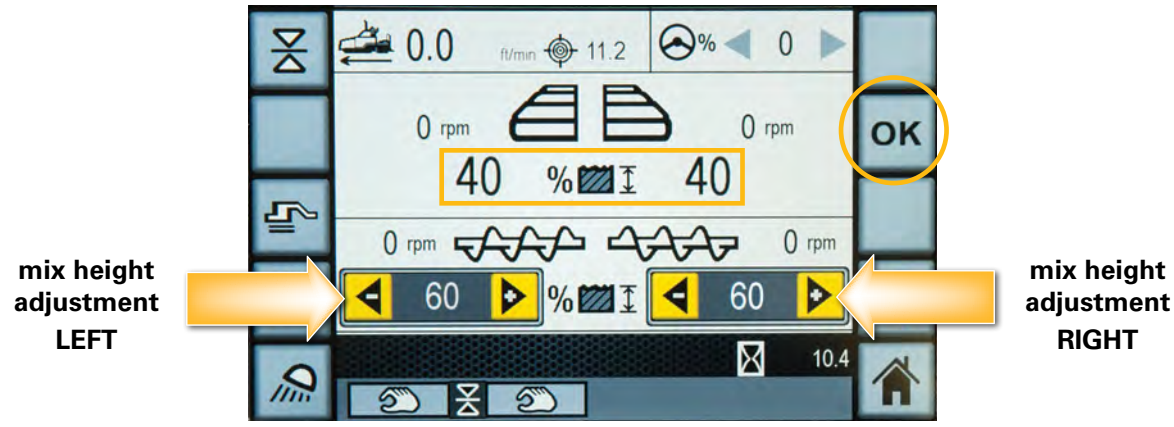
Use the control panels to set feeder system controls.

MATERIAL FEED ICON



Select the material feed adjustment icon on the left side of the tractor control panel touch screen.

MATERIAL FEED CONTROL SCREEN



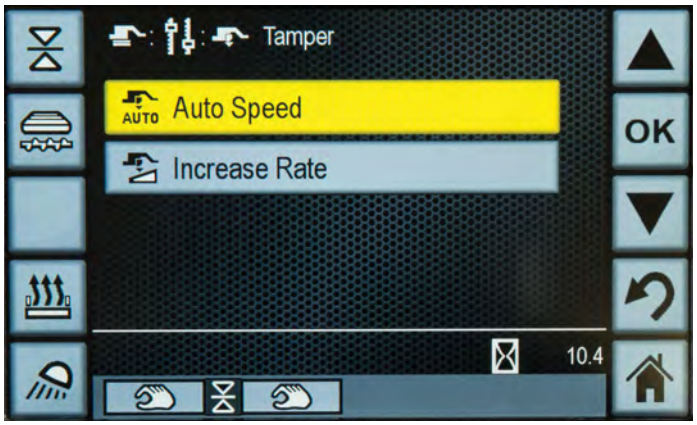
The feeder control sets conveyor speed and mix height.

Using the multi-function dial on the control panel, set the conveyor speed to 40 percent for each conveyor. Adjust the mix height to 60 percent using the arrows on the touch screen. After adjusting mix height, push the OK icon on the touchpad to accept the setting. This is a starting point for the feeder system. The tractor operator or screed operator can fine tune these settings when paving begins.

Use the screed control panel to set tamper and vibration speeds.



On the main screed control panel, set the tamper bar speed to 1200 RPM using the multifunction dial. Set the tamper bar ramp time control to the minimum position. Set the vibration frequency control to 1500 RPM.



Depending on the type of mix and the paving speed, adjust each control when paving starts. The objectives are to achieve the target density for pre-compaction by the screed and to produce a mat with consistent texture and smoothness. As a rule, a higher paving speed requires a higher tamper speed and higher vibration frequency. The tamper ramp time is normally adjusted when using stabilized material for base. When laying down bituminous material, the tamper ramp time is set at "Minimum," so that it starts immediately when paving begins.

[12. FILL THE AUGER CHAMBER]

Filling the auger chamber to the correct level is extremely important. The most common mistake is overfilling the auger chamber. Remember, the material in the auger chamber is weight, or resistance, felt by the screed. If there is too much resistance, the screed will climb at take-off and create a high spot at the transverse starting joint.



Fill the auger chamber using manual conveyor and auger controls.

Use the left and right conveyor controls at the operator station to fill the auger chamber.

User Tip: *Always fill the auger chamber with the engine at low idle speed. At low idle, the hydraulic drive for the feeder system operates slowly. The slower conveyor speed minimizes the amount of material that gets pushed under the tractor. Also, slower auger speed minimizes segregation when paving with a mix containing large-size aggregates. It is easier for the operator to control the flow of material across the entire auger chamber using the left and right augers.*

Start with either the left or right conveyor. Hold the manual control down and slowly fill the auger chamber behind that conveyor until the material covers the augers in the center.



Manual auger controls are located below the conveyor controls.

Select the manual auger control for that side and depress the control until all material has been moved out as far as possible.

Alternately, use the manual conveyor control and the manual auger control to fill the auger chamber and to create a head of material that covers one half of the auger shaft or slightly more. Repeat the procedure for the other side of the auger chamber.

Use a shovel to move material out to the end gate whenever the extender is beyond 30 cm (1') away from the last auger segment or whenever special attachments are added to the end of the screed. Using the auger to force mix out to a wide paving width will create an overload in the center of the auger chamber and the screed will climb during takeoff.

Do not fill in the area adjacent to the main screed and directly in front of the screed extender. This area will be filled by material as the paver pulls forward off the starting reference.



Use a shovel to move material out to the end gate.



Put feeder controls in the Automatic mode.

When the auger chamber has been filled half full—no more, no less—set the conveyor and auger mode controls on the console to Auto. Either push the automatic conveyor and auger button or press individual automatic buttons for both the conveyor and auger. Verify that all four automatic system indicators are illuminated. If the controls remain in the Manual mode, neither the conveyors nor the augers will operate when paving operation begins.



[13. SET ACCESSORY FUNCTIONS]

Refer to the Operations and Maintenance Manual to review the list of available accessory functions. Access the accessory functions by touching the Machine Functions icon. Some of the accessory functions available are:

- Friction Steer
- Vibrator System On/Off and Speed
- Screed Assist System
- Timers
- Propel Hill Hold
- Automatic Engine Speed Control



Before pulling off the starting reference, set accessory functions.



Set the Grade Control system before pulling off the starting reference.

Automatic grade and slope controls are considered accessory controls. Set them according to the job specifications. Install any averaging devices and/or grade sensors as required. Select either single-side screen display or dual-side screen display per operator preference.

[14. PULL OFF THE STARTING REFERENCE]



*Make sure the gear selector is in the PAVE mode **(1)**, the screed is in float **(2)**, and the engine throttle is set at the chosen idle speed (ECO or high idle) **(3)**. Then, turn the speed control dial **(4)** clockwise until the calculated paving speed is displayed in the target speed area of the display **(5)** and release the parking brake **(6)**. Propel the machine by squeezing the trigger and stroking the propel lever **(7)** fully forward in one quick, smooth motion.*

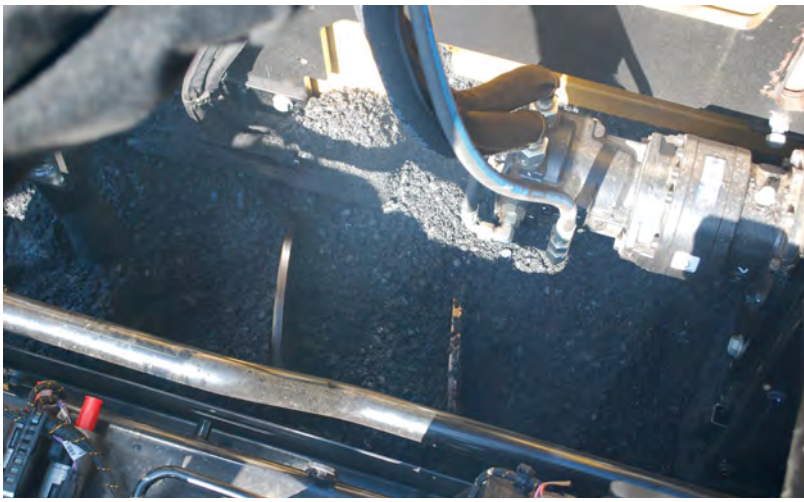
***Note:** When the calculated target speed is reached, use the propel lever to stop and start the machine using the same quick, smooth motion as before. Slowly stroking the propel lever will result in a more uneven head of material and have an adverse effect on mat smoothness.*

The left and right screed operators should be monitoring the area at the end of the auger shaft and in front of the extender. The primary responsibility at the start of paving is to ensure that there is enough material coming out of the auger chamber to flow to the end gate. The screed operator can adjust the mix height dial to set the head of material at the end of the auger shaft and to set the flow of material in front of the extender.



The screed operators are responsible for making sure enough material flows to the front of the extenders.

Next, the paver operator checks the level of material in the center of the auger chamber. Use the left or right ratio control dials to adjust the head of material. Once the head of material is correct across the full width of the auger chamber, the auger rotational speed must be monitored and adjusted if necessary.



Observe the head of material in the center of the auger chamber.

User Tip: *On many Cat Pavers, the speed of the left and right augers is shown on the LCD displays. On older pavers without auger speed indicators, the operator can simply monitor the rotation of the last auger segment. The last auger segment is a split segment. As the tip of the segment rotates, count how long it takes for one complete revolution. A two-second revolution equals 30 revolutions per minute. Remember, the ideal range for auger speed is 20 to 40 revolutions per minute.*

To increase auger speed, turn the ratio control dial counterclockwise. This slows the speed of the affected conveyor and reduces the volume of material in the auger chamber. Hence, the sensor will signal for an increase and the augers will rotate more quickly. To slow the auger rotation, turn the ratio control dial clockwise.

[15. OPERATIONAL ADJUSTMENTS]



A good takeoff should result in uniform appearance of the asphalt layer.

The screed will be at its equilibrium point when the paving speed has stabilized and the feeder system is operating at a uniform speed with the correct head of material. Forces acting on the screed are in balance and the screed is now floating at a constant level. Check and adjust layer thickness and slope and also check the appearance of the layer. If screed height, angle of attack and slope are correct, the mat should have a uniform texture across the full width of the mat, and no longitudinal lines should be present.



Extender height too high

If an extender is set too high, a longitudinal transition mark in line with the outer edge of the main screed will appear. The mat will be thicker behind the screed extender. First try decreasing the tamper speed. This will cause the screed to lower, decreasing mat thickness. The tow-points are then adjusted up increasing the angle of attack returning to the original mat thickness. When the angle of attack is increased, the separation marks will decrease. Continue to adjust the tamper speed until the marks disappear.

Lower the extender until the transition marks are eliminated.

Note: *If the extenders were not calibrated to the main screed, mechanical adjustments may be required to completely remove the separation marks.*



Extender height too low

If a slight mismatch occurs between the height of the extender and the height of the main screed a line will appear longitudinally in the surface of the layer.

If an extender is set too low, a longitudinal transition mark in line with the inner edge of the screed extender will appear. A thinner mat behind the screed extender will be evident. First try increasing the tamper speed. This will cause the screed to lift, increasing mat thickness. The tow-points are then adjusted down, decreasing the angle of attack returning to the original mat thickness. When the angle of attack is decreased, the separation mark will decrease. Continue to adjust the tamper speed until the mark disappears.

Lower the extender until the transition marks are eliminated.

Note: *If the extenders were not calibrated to the main screed, mechanical adjustments may be required to completely remove the separation marks.*

Two of the most important fundamentals of smooth paving are maintaining a consistent paving speed and a constant head of material.

Normally, the paver can pave smoothly at any speed that matches the delivery of mix to the worksite. The important thing is to keep the speed constant.

If the paving speed is changed drastically, the screed will either rise or fall, and mat smoothness will suffer.

Also, if paving speed changes, the demands on the feeder system also change and the feeder system must be adjusted to match the new material demands.

For quality paving results, always follow the basic fundamentals of paver set-up and keep the operation consistent.



[SUMMARY]

The proper takeoff from the starting reference sets the tone for the length of the pass. The crew should take the time to do every startup step correctly and in the proper sequence.

Two steps are particularly important. The first is setting the main screed angle of attack. Be sure the crew knows the characteristics of the screed with which they're working. What they learned on one screed may be different than what is needed on another screed. Remember, the angle of attack that is installed while the screed is setting on the starting reference is the angle of attack the screed returns to for the length of the pass.

Second, fill the auger chamber slowly, one side at a time. It is very common for the crew to overfill the auger chamber at the starting point. Always fill with the engine at low idle. It takes a few minutes longer to slowly fill the auger chamber, but it pays off in smoother transverse joints.

Finally, be sure the crew is paving with a calculated paving speed whenever possible. If the speed changes, adjust the feeder system to match the new demand for material.


GRADE CONTROL
AP555F
CAT



Unit 6
AUTOMATIC GRADE AND SLOPE CONTROL

Automatic grade and slope control is designed to make the crew's jobs easier and improve the quality of the final product. A crew that is well-trained in the uses of grade and slope control can set the paver up to maximize productivity and paving efficiency in a wide range of applications.



Unit 6 provides information about automatic grade and slope control, one of the four basic elements to quality construction of asphalt layers. Several types of grade and slope systems are available on pavers. The components shown in this unit are primarily from the Cat Grade Control System and the Trimble Paving Control Systems. Other systems may differ, however, the basics of grade and slope setup and operation are the same for all systems.

Grade control can be thought of in several ways. On some projects, it can be considered the control for thickness of the layer being placed. Many projects require the layer thickness to remain constant so the yield (the number of tons calculated for a project versus the number of tons actually laid down) is exactly what has been planned.

On other projects, grade control may be thought of as the control for elevation of the layer that is being placed. On that type of project—airport runways, for example—the layer thickness can vary in the quest to create exact elevations.

Grade control can be used to improve the longitudinal profile (smoothness or ride quality) of the bituminous structure. In the case of paving for best ride quality, grade control ignores thickness and elevation and concentrates on eliminating roughness. These applications will be reviewed in detail in this unit and in Unit 7.

Slope control creates the specified transverse profile (slope). When selecting automatic slope control on one side of the paver, thickness and ride quality may be compromised.

Automatic grade and slope control will not correct mistakes that the crew makes. Automation can make the crewmembers' jobs easier, but the crew must always follow the fundamentals of paving and use best practices whether they are paving manually or using automation.

[GRADE AND SLOPE COMPONENTS]

Grade and slope control systems consist of one or more grade sensors, a slope sensor, a control module, and display units. Main system power cables, grade sensor cables and slope sensor cables are also utilized. On some pavers, many of the components are built into the tractor and screed and stay on the paver at all times. On other pavers, most grade and slope components are installed on the paver at the start of every shift and removed for storage at the end of the shift. Always follow the manufacturer's guidelines for installation and storage of grade and slope system components.

User Tip: *The main power cable from the tractor to the control panel or to the system display carries 24 volts. Before connecting the power cable to the control panel or to the display, be sure to have the paver key switch in the OFF position. If there is power in the cable, it is possible to cause damage to the cable or to the control panel or display when connecting the cable.*

[SONIC GRADE SENSORS]

Grade sensors are either sonic or mechanical.

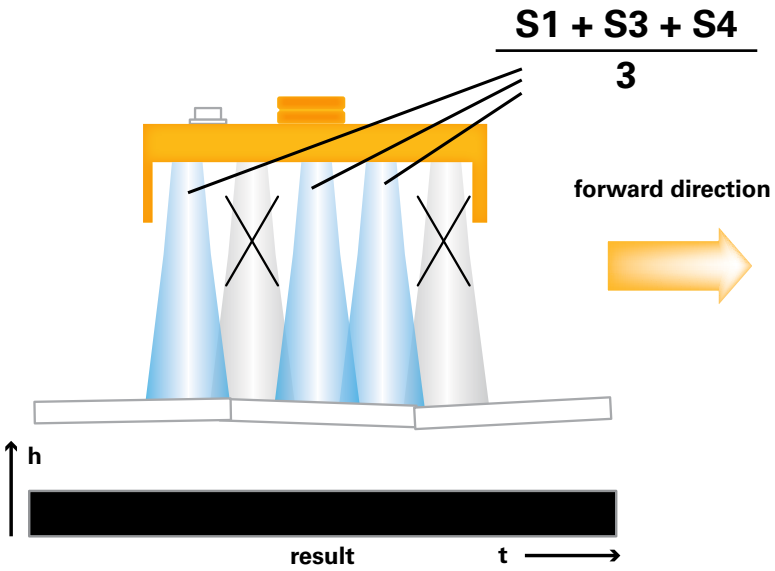
Sonic sensors do not touch the grade reference. Instead, sonic sensors emit sound pulses. The sound pulses travel from the sensor to the grade target. The grade target can be an existing roadway surface, a milled surface, an adjacent bituminous layer, a gutter, or a string. The sound pulses rebound from the grade reference back to the sensor.

When the grade sensor emits a sound pulse, a timer starts. When the echo returns to the sensor, the timer stops and this information is continuously sent to the control panel or to the system controller. The speed of sound is a known factor. The controller uses the time it takes for the sound pulses to rebound back to the sensor to calculate the distance between the sensor and the grade target.

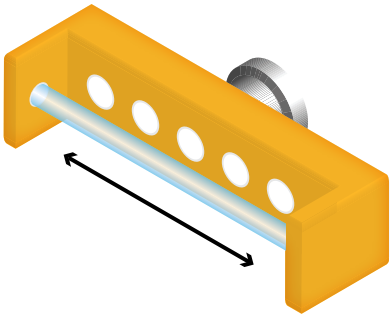
Some sonic sensors have a single transducer that emits one set of sound pulses. Cat Sonic Sensors have five transducers.



Sonic sensors use sound pulses to measure the profile of the grade over which the bituminous layer is being placed.



The Cat Sonic Sensor makes five grade measurements.



On Cat grade sensors, a transducer emits horizontal sound pulses to detect rapid air temperature changes.

When using a Cat Sonic Sensor to read any grade reference other than a stringline, install the sensor so it is parallel to the direction of paving with the cable connection toward the rear of the paver. The five transducers make five separate measurements. The system ignores the two measurements that have the most deviation from the benchmark distance. The other three measurements are averaged. The average of the three best measurements is sent to the control module.

Sound pulses travel faster through hot air than through cooler air. The grade sensor senses ambient temperature and corrects the distance measurements it sends to the controller based on the ambient air temperature.

However, the grade sensor may be exposed to rapid changes in air temperature on a project. Therefore, grade sensors are also equipped with devices to detect rapid changes in air temperature.

Cat grade sensors have a built-in system to detect rapid air temperature changes. A transducer is built into a tab that extends about 10 cm (4") below the transducers that emit the grade-sensing sound pulses. That transducer emits horizontal pulses that rebound off the tab on the other side of the grade sensor. The speed of the sound pulses across this known distance is used to compensate for rapid air temperature changes. Other grade sensors use detachable temperature bails that serve the same purpose.

Sound pulses rebound back to the grade sensor off the first item that is within the range of the sound pulses. Therefore, the crew should take precautions to keep the grade reference area free of all objects.